Shlok Lande

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Education

University of British Columbia Vancouver, BC

Bachelor of Applied Science in Computer Engineering, Co-op Program *Sep 2025 – May 2029*

- **Awards:** Dean's List

Skills

- **Languages:** C, C++, Python, Java, JavaScript, TypeScript, System Verilog, ARM Assembly
- **Web & Backend:** React, Next.js, React Native, Node.js, Supabase, Tailwind
- **Embedded:** FreeRTOS, I2C, SPI, UART, CAN bus (DBC), oscilloscope/logic analyzer debugging
- **Hardware:** Quartus Prime, ModelSim, DE1-SoC board, STM-32 Board
- **Cloud & Tools:** AWS, Docker, Git, Linux, InfluxDB, Grafana, JUnit, Gradle

Experience

UBC Solar May 2026 – Present

Embedded Software Developer *Vancouver, BC*

- Wrote SHTP/SH-2 host protocol from scratch in C to bring up a BNO086 IMU over I2C, validating on a bare-metal bench build before FreeRTOS integration
- Developed FreeRTOS IMU firmware packing orientation, acceleration and magnetic field data into CAN frames at 100 Hz for the solar car's real-time telemetry bus transmitting CAN messages over the air via radio and cellular
- Diagnosed I2C bus lockups during sensor bring-up with a logic analyzer and wrote a bus-recovery routine, eliminating hangs that previously required manual power-cycling
- Configured independent watchdogs (IWDG) across 5 onboard boards, enabling automatic recovery from firmware hangs without manual intervention
- Added a CAN bus-load metric reporting live utilization so new messages can be scheduled without saturating the [500 kbps] bus
- Created Grafana dashboards over InfluxDB with Flux queries, streaming live CAN signals to pit crew during testing

Projects

match-a — Full-Stack Volunteer Matching Platform | *youCode 2026 — 2nd Place, SAP Non-Profit Category*

- Shipped a two-sided volunteer-matching platform for BC nonprofits in under 24 hours with a team of three, with **Supabase Realtime WebSockets** powering live dashboards
- Trained a **logistic regression** churn model in Python to flag volunteers at risk of dropping off
- Enforced role-based access control via **Supabase Auth**, giving volunteers and coordinators separate dashboards for impact tracking and organization management
- Designed an urgent-request system pushing coordinator crisis alerts to volunteers in real time, plus interest-based onboarding matching new sign-ups to opportunities
- *Website:* match-a-seven.vercel.app

COMMUTEity — AI-Powered Commute Matching App | *AWS Hackathon*

- Architected **AWS** backend, standing up Cognito auth, DynamoDB and a Lambda-based matching pipeline
- Designed a geohash-based location model, with a Lambda overlap engine comparing commute sessions pairwise to detect recurring overlap between users
- Integrated **Amazon Bedrock** (Claude Sonnet 4.5) to turn raw overlap statistics into natural-language match explanations

Code Unity — Co-Founder & Vice President *Non-Profit Organization*

- Co-founded a nonprofit teaching introductory programming, delivering 24 weekly lessons on algorithmic thinking to an average of 10 students
- *Website:* codeunity.wixsite.com